CS 5361/6361 Machine Learning
Spring 2026

1. General Information

Instructor:

Olac Fuentes
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Web: www.cs.utep.edu/ofuentes
Office hours: Tuesdays and Thursdays 2:00 pm - 3:30 pm in CCSB 3.0412 or by appointment
Feel free to contact me on MS Teams at other times
Note: right before or right after class are not good times to contact me

Meeting times and place:
Mondays and Wednesdays 1:30 pm - 2:50 pm in CCSB 1.0202

Catalog Description:
Machine learning studies the development of programs that can improve in the performance of a task with experience. For many difficult problems, such as speech understanding, image classification, and text analysis, solutions based on machine learning outperform all others proposed to date. In this course we will study several of the most commonly used machine learning algorithms, their application to problems in different areas of interest, and their quantitative evaluation. We will also discuss current research issues in machine learning. Each student will do a research project related to a problem of his/her interest.

Outcomes:
On successful completion of this course, students will:

1. Understand basic machine learning tasks, including regression, classification, and clustering.
2. Evaluate the performance of machine learning models.
3. Learn representative examples of each of the following classes of algorithms: instance-based methods, decision trees, kernel-based methods, probabilistic approaches, ensembles, and neural networks.
4. Apply data preprocessing and feature selection techniques.
5. Implement machine learning algorithms in Python.
6. Use machine learning libraries such as Sklearn and Tensorflow.
7. Apply machine learning algorithms to solve real-world problems.

Master's Level vs. Doctoral Level Expectations:
Doctoral level students are given additional and more advanced assignments and exam questions. The course includes a final project; for undergraduate students, implementation of well-known algorithms is usually sufficient, for graduate students, projects are expected to include advanced algorithms, analyses, and/or applications.

Extended Project Option:
Students who have a good background in the area and are interested in research may opt for a semester-long individual project. Stop by my office during office hours if you are interested in this option.

Prerequisites:
Ability to write non-trivial programs in Python/Numpy
Calculus, Matrix Algebra, and Probability and Statistics

2. Course Contents (tentative)

1) Introduction
    a) Machine Learning: definition and motivation
    b) Machine Learning tasks: classification, regression, clustering, reinforcement learning
    c) Review – Python and arrays
    d) Review – Linear algebra and probability
2) Performance evaluation
    a) Tasks and metrics
    b) Cross-validation
        i) Training and test sets
        ii) N-fold cross validation
        iii) Leave-one-out cross validation
3) Data preprocessing
    a) Normalization
    b) Standardization
    c) Principal component analysis
    d) One-hot encoding
    e) Feature selection
    f) Feature creation
    g) Data augmentation
4) Instance-based learning
    a) K-nearest neighbors
5) Linear regression
6) Logistic regression
7) Statistical methods
    a) The Naïve Bayes algorithm
8) Support vector machines
9) Tree-based algorithms
    a) Decision trees
    b) Regression trees
10) Ensembles of models
    a) Random forests
11) Neural networks
    a) Feed-forward dense neural networks
    b) Convolutional neural networks
12) Image classification
13) Text classification
    a) Bag-of-words representation
    b) Word embeddings

3. Policies and Other Information

Textbook:
There will be no textbook. We will use Colab notebooks and other online resources.

Grading:
Lab assignments, homework, and quizzes: 30%
Partial exams (2): 30%
Final Exam (comprehensive): 20%
Final Project: 20%

Note: in order to pass the course, students must have an average of at least 50 in the partial exams and a grade of at least 50 in the final exam.

Policy on missed assignments and quizzes: Since results are posted immediately, it will not be possible to make up for missed quizzes and homework, even if they are missed for justifiable reasons. I will ignore the two lowest homework and quiz grades to help minimize the possible negative effects from this policy.

Policy on attendance: Students are expected to attend all lectures. A student missing more than three lectures without making prior arrangements may be dropped from the class. Students arriving more than five minutes late will not be allowed into the classroom.

Requests:

- Bring your laptop and paper and pencil to every lecture
- Do not open your laptop during class unless we are working on an exercise
- Attach a label with your name to the back of your laptop
- Turn on your camera for remote meetings

Disabilities: If you feel that you may have a disability that requires accommodation, contact the Center for Accommodations and Support Services (CASS) (https://www.utep.edu/student-affairs/cass/).

4. Standards of Conduct and Academic Dishonesty

Students are expected to maintain absolute integrity and a high standard of individual honor in scholastic work undertaken at the University. Professors are required to report academic dishonesty and any other violations of the Standards of Conduct to the Dean of Students.